# Systemadministration

Introduktion

## Filsystem

# Agenda

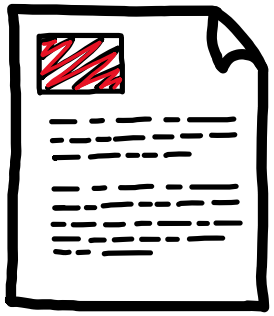
En introduktion till filsystem

1. Filsystem & sekundärminne
2. Datakorruption
3. Träd
4. Filstruktur
5. Navigering med CLI
6. Filsystem
7. NTFS filrättigheter

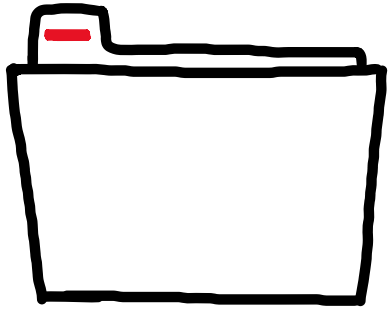
# 1. Filsystem & sekundärminne

# Filsystem

Ett filsystem består utav:



**File**
**Fil**



**Directory, folder**
**Katalog, mapp**

# Olika typer av sekundärminne

Fungerar som ett ’långtidsminne’

- **Magnetband**
- **Hårddisk**
- **Flash-baserad lagring**

# Magnetic Tape

Magnetband

Utvecklades 1928 och har använts för bland annat:

Kassettband, VHS…

Idag allt mer ovanligt, används främst för **arkivering** och **backup**

**LTO**

*LTO 2. generatsiooni lindiseadme kassett*, by T2hep, 2016, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=48407033). CC BY-SA 4.0.

# **H**ard **D**isk **D**rive
## Hårddisk

Introducerades 1956 och har varit de facto standard för *secondary storage*

*A pair of Seagate Barracuda hard drives*, by Ashley Pomeroy, 2020, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=93199385). CC BY-SA 4.0.

# Flash Memory

Det finns flera typer av **flash**-baserad lagring

Har historiskt varit för dyrt och/eller opålitligt (data försvinner)

*A SanDisk-brand USB thumb drive, SanDisk Cruzer Micro, 4GB*, by Evan-Amos, 2019, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=11294374). Public domain.

# **I**nput/**O**utput

- Sekundärminnen är **I/O**–enheter
- De förser processor & primärminne med data
- Processorn **läser** (**read**) eller **skriver** (**write**) data med **I/O operations** till/från sekundärminnet

# Input/Output

*Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam.* by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

Beräkningar

*MT36JSF1G72PZ-1G9K1HE; Memory Type DDR3 SDRAM; Capacity 8GB; Data Transfer Rate 1866MHz; Pins 240; Bus Type PC-14900; Error Correction Registered ECC; Cycle Time 1.07ns; Cas CL13; Memory Clock 233MHz; Voltage 1.5V; Manufacturer Micron Technology, Inc.*, by Mister rf, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=129056892). CC BY-SA 4.0

I/O operations

read/write

# Volym

Volume

En lagringsyta med ett installerat filsystem

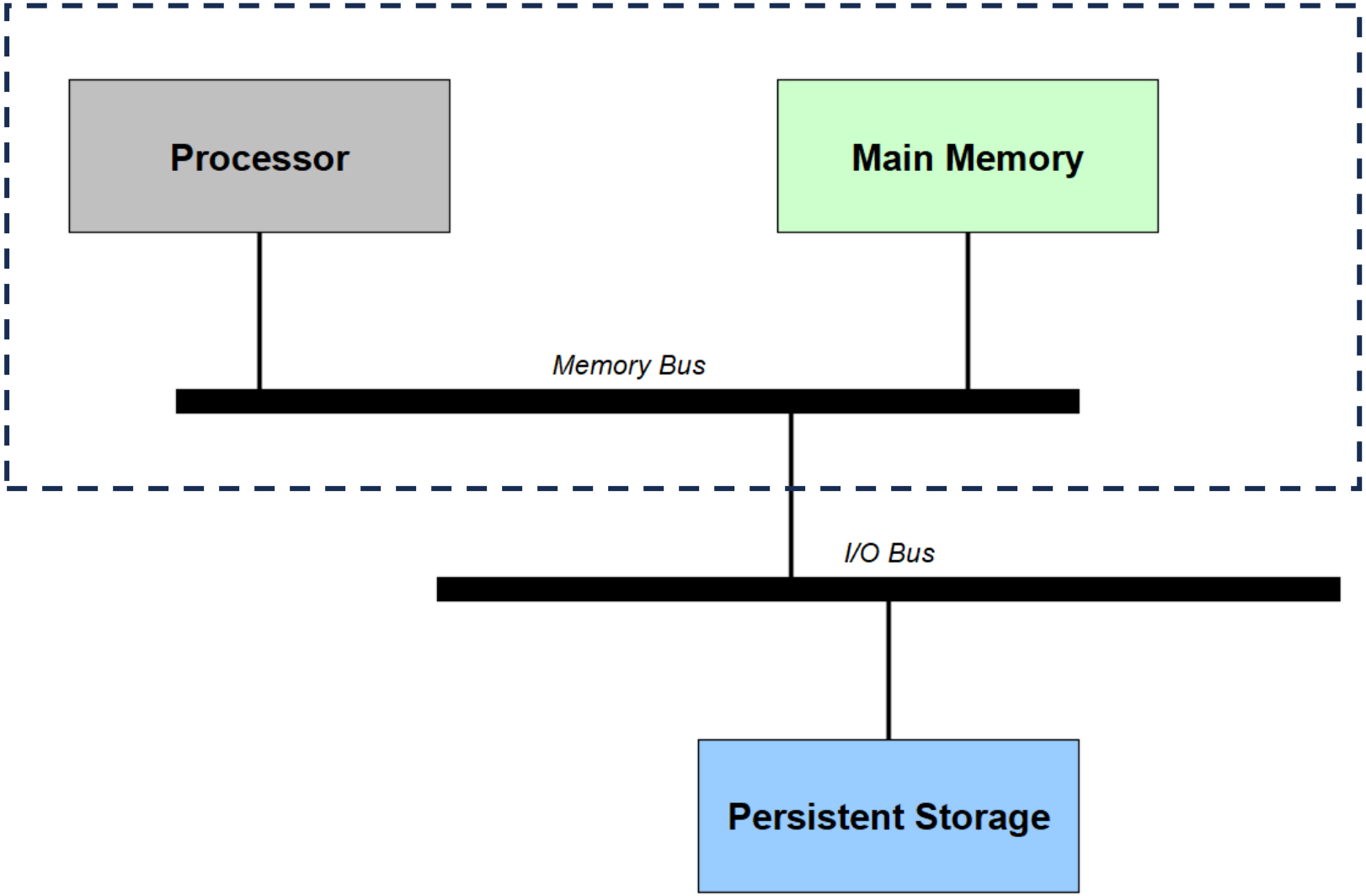
Ett filsystem + sekundärminne = volym

# 2. Datakorruption

Snabbt

*Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam.* by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

*MT36JSF1G72PZ-1G9K1HE; Memory Type DDR3 SDRAM; Capacity 8GB; Data Transfer Rate 1866MHz; Pins 240; Bus Type PC-14900; Error Correction Registered ECC; Cycle Time 1.07ns; Cas CL13; Memory Clock 233MHz; Voltage 1.5V; Manufacturer Micron Technology, Inc.*, by Mister rf, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=129056892). CC BY-SA 4.0




Långsamt

*Top view of an Intel central processing unit Celeron Kaby Lake type core, model G3930, launched in 2017. LGA 1151 socket, 14 nm process, core frequency 2.90 GHz, Thermal Design Power 51 W, dimensions 37.5mm x 37.5mm. Manufactured in Vietnam.* by Erica Gaba – Wikimedia Commons user: Sting, 2022, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=123062267). CC BY-SA 4.0

Beräkningar

Snabbt

*MT36JSF1G72PZ-1G9K1HE; Memory Type DDR3 SDRAM; Capacity 8GB; Data Transfer Rate 1866MHz; Pins 240; Bus Type PC-14900; Error Correction Registered ECC; Cycle Time 1.07ns; Cas CL13; Memory Clock 233MHz; Voltage 1.5V; Manufacturer Micron Technology, Inc.*, by Mister rf, 2023, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=129056892). CC BY-SA 4.0

**Processor**
‘hjärnan’

**Primärminne**
‘arbetsminne’

Läs/
Skriv

Långsamt

I/O buffras

Processor & primärminne utför andra beräkningar under tiden

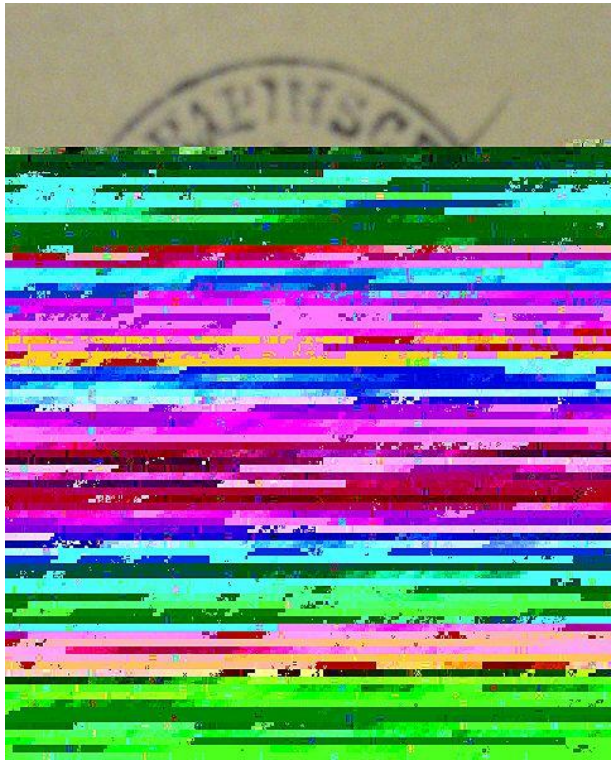
**Sekundärminne**
‘långtidsminne’

# Datakorruption

Data förvrängs vilket leder till att information går förlorad

Kan ske av flera anledningar…

Vanligt när skrivningar till filsystem plötsligt avbryts



*Result of an almost completely failed data restoration of a JPEG photograph on a hard drive. The file was overwritten after about a year with other data and is visible only to a small extent in the original,* by Mattes, 2008, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=5217861). Public domain.

# Äldre system var känsligare…

## Ex. Hur man bör stänga av ett UNIX system

```
root@UNIX:~$ echo 'shutting down system for maintenance' | wall
shutting down system for maintenace
root@UNIX:~$ w
USER        TTY         FROM              WHAT
root        tty0                          w
hans        pts/3       172.0.2.18        nano notes
root@UNIX:~$ pkill –u hans
root@UNIX:~$ lsof
COMMAND     PID         USER        NAME
ftpd        1248        root        /srv/ftp
root@UNIX:~$ kill 1248
```

# Äldre system var känsligare…

## Ex. Hur man bör stänga av ett UNIX system

```
root@UNIX:~$ kill 1248
root@UNIX:~$ sync
root@UNIX:~$ sync
root@UNIX:~$ sync
root@UNIX:~$ shutdown -y -g 15 -i s 'system going down'
system is going down
```

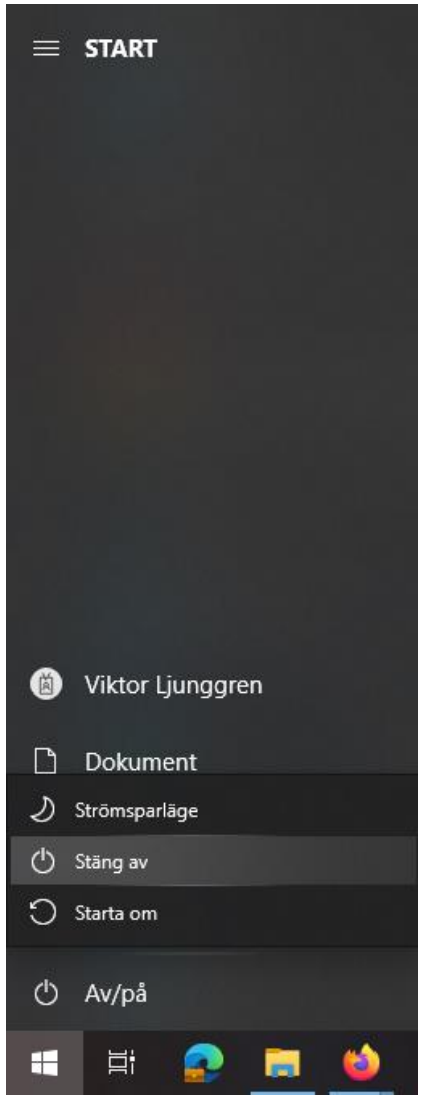
“[...] one of the important parts of the shutdown process is syncing the disks. The `sync` command finishes all disk transactions and writes out all data to disk, guaranteeing that the system can be turned off without corrupting the files [...]”

Frisch, Æ. (2002). *Essential System Administration* (3rd ed.) (pp. 172). O’Reilly Media.

”Why is `sync` executed two or three times (or even more)? [...] The command schedules but does not necessarily immediately perform the required disk writes [...] Typing “`sync`” several times gives you something to do while you’re waiting”

Frisch, Æ. (2002). *Essential System Administration* (3rd ed.) (pp. 173). O’Reilly Media.

# Shutdown



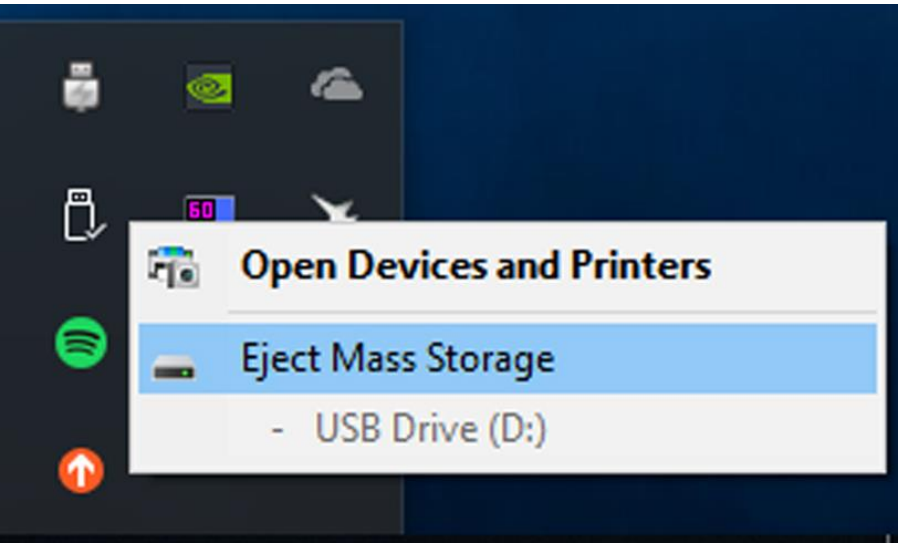

Modern **shutdown** löser det åt dig

Men om du inte använder en **graceful shutdown** för att stänga av kan samma problem uppstå

# Do You Safely Remove USB Flash Drives?

Du kan faktiskt ta bort USB-minnet utan att ’mata ut den säkert’ först…

Och så länge inget skrivs till den under tiden så ska datakorruption inte uppstå…



*A SanDisk-brand USB thumb drive, SanDisk Cruzer Micro, 4GB*, by Evan-Amos, 2019, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=11294374). Public domain.

# 3. Träd

# **Träd**

Tree

Rötter, grenar, löv…

*Green and yellow tree under blue sky during daytime*, by Ivan Rohovchenko, 2019, Unsplash (https://unsplash.com/photos/p3vrfHDr46s). Unsplash license.

# Träd

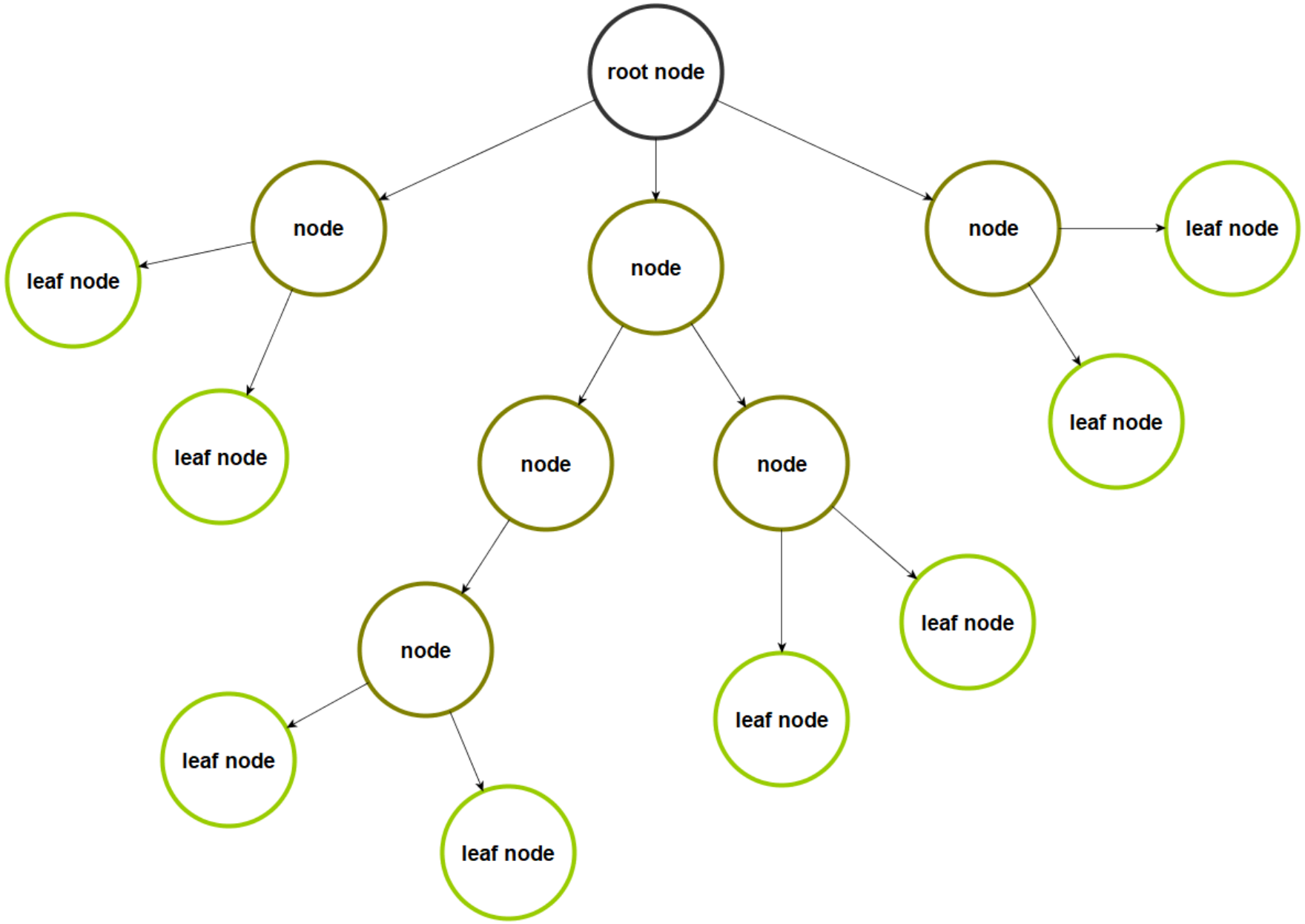
Tree

Datorers träd är lite mer abstrakta…

*Green and yellow tree under blue sky during daytime*, by Ivan Rohovchenko, 2019, Unsplash (https://unsplash.com/photos/p3vrfHDr46s). Unsplash license.

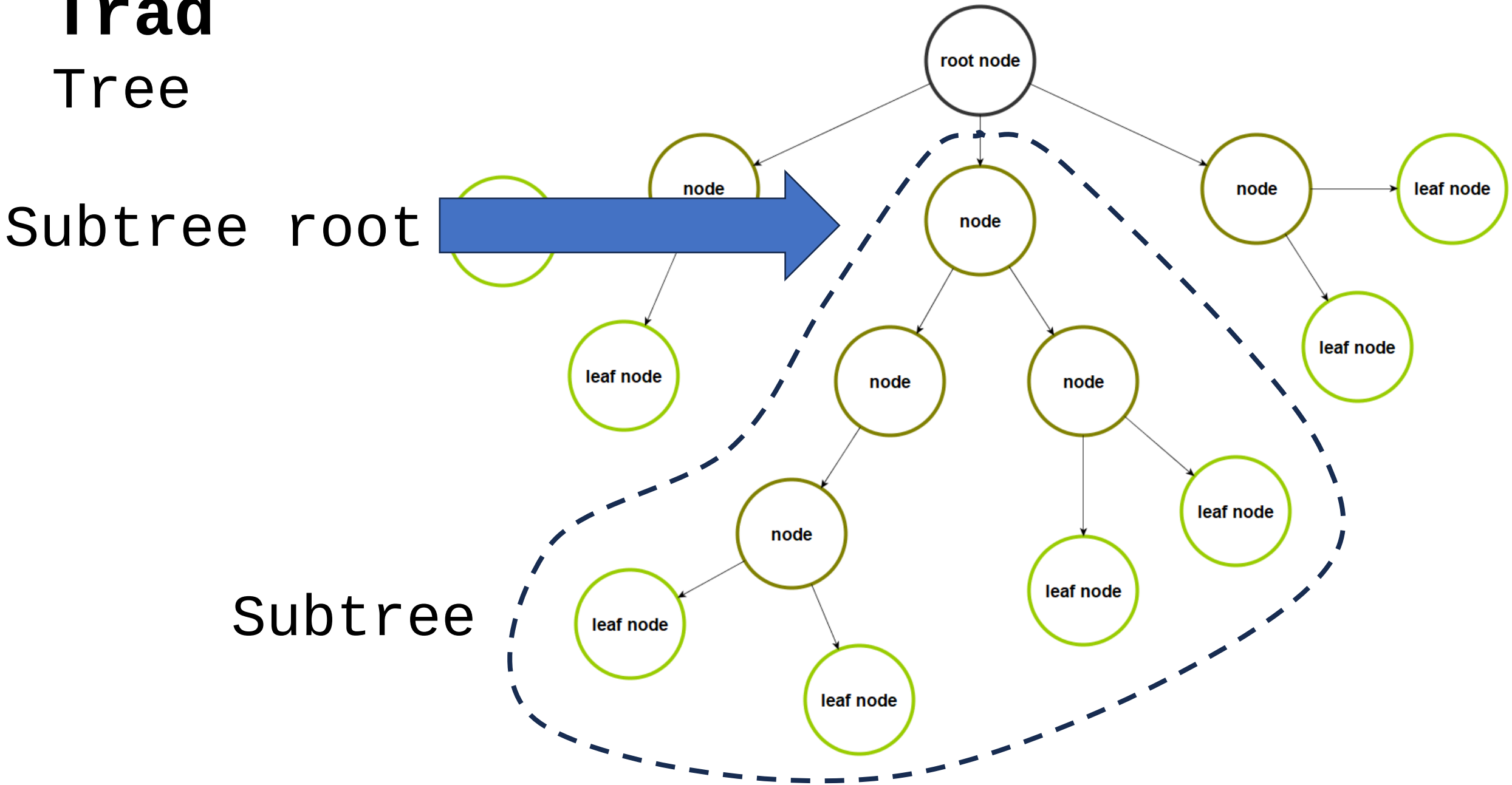
# Träd

Tree


root node
node
node
node
leaf node
leaf node
leaf node
leaf node
node
node
leaf node
node
leaf node
leaf node
leaf node

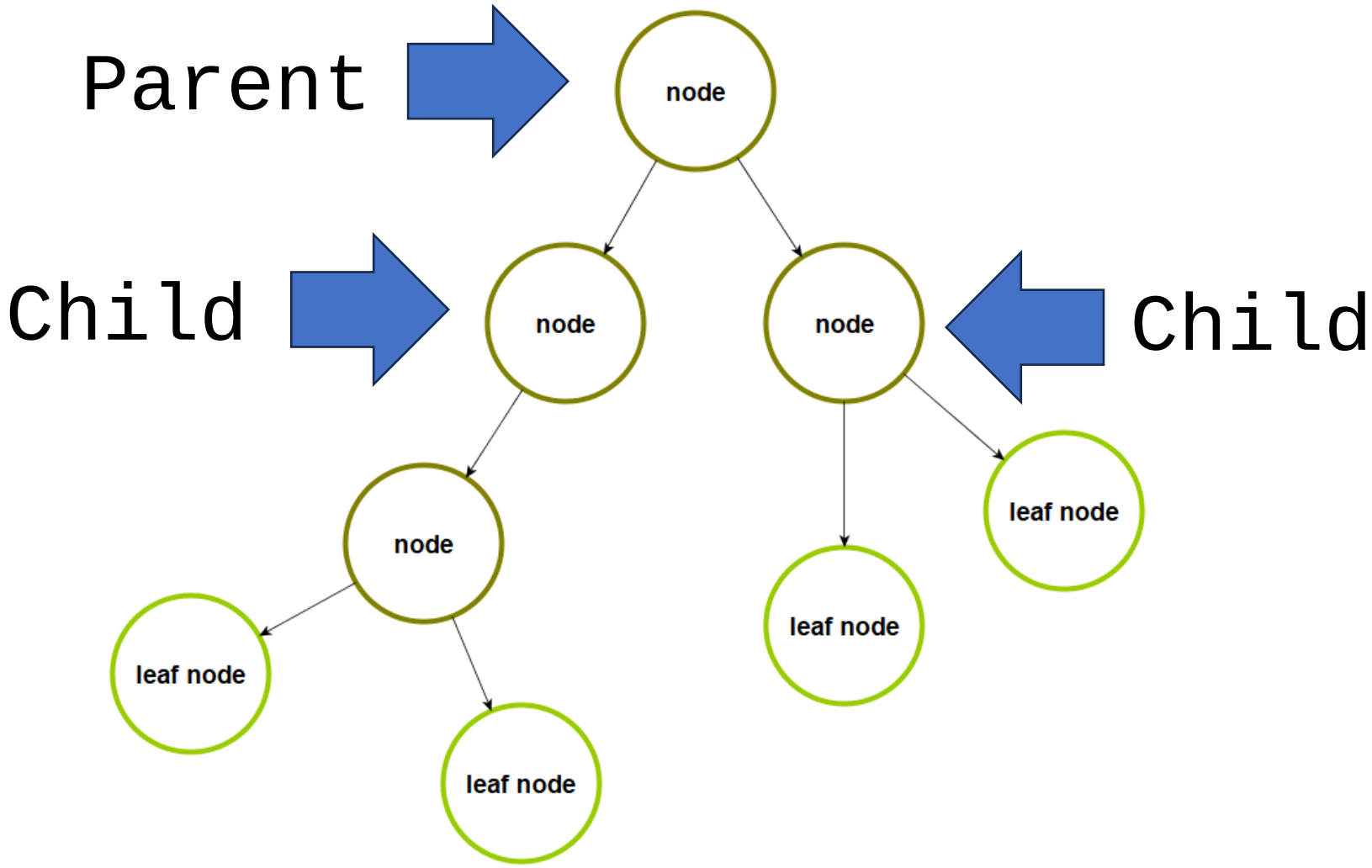
# Träd

Tree


Subtree root
Subtree
root node
node
node
node
leaf node
leaf node
leaf node
node
node
leaf node
node
leaf node
leaf node
leaf node

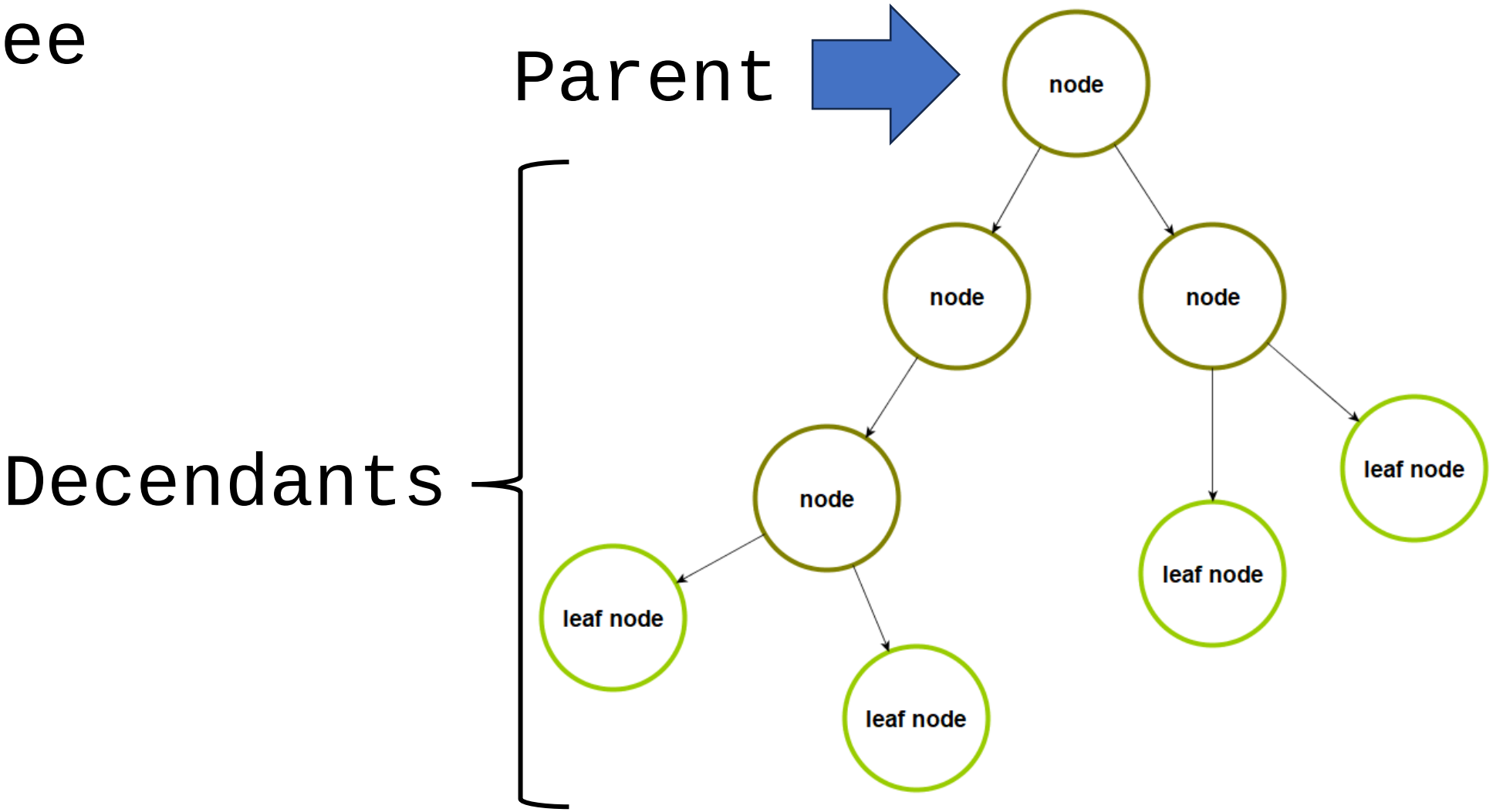
# Träd

Tree


Parent
node
Child
node
node
Child
node
leaf node
leaf node
leaf node
leaf node

# Träd

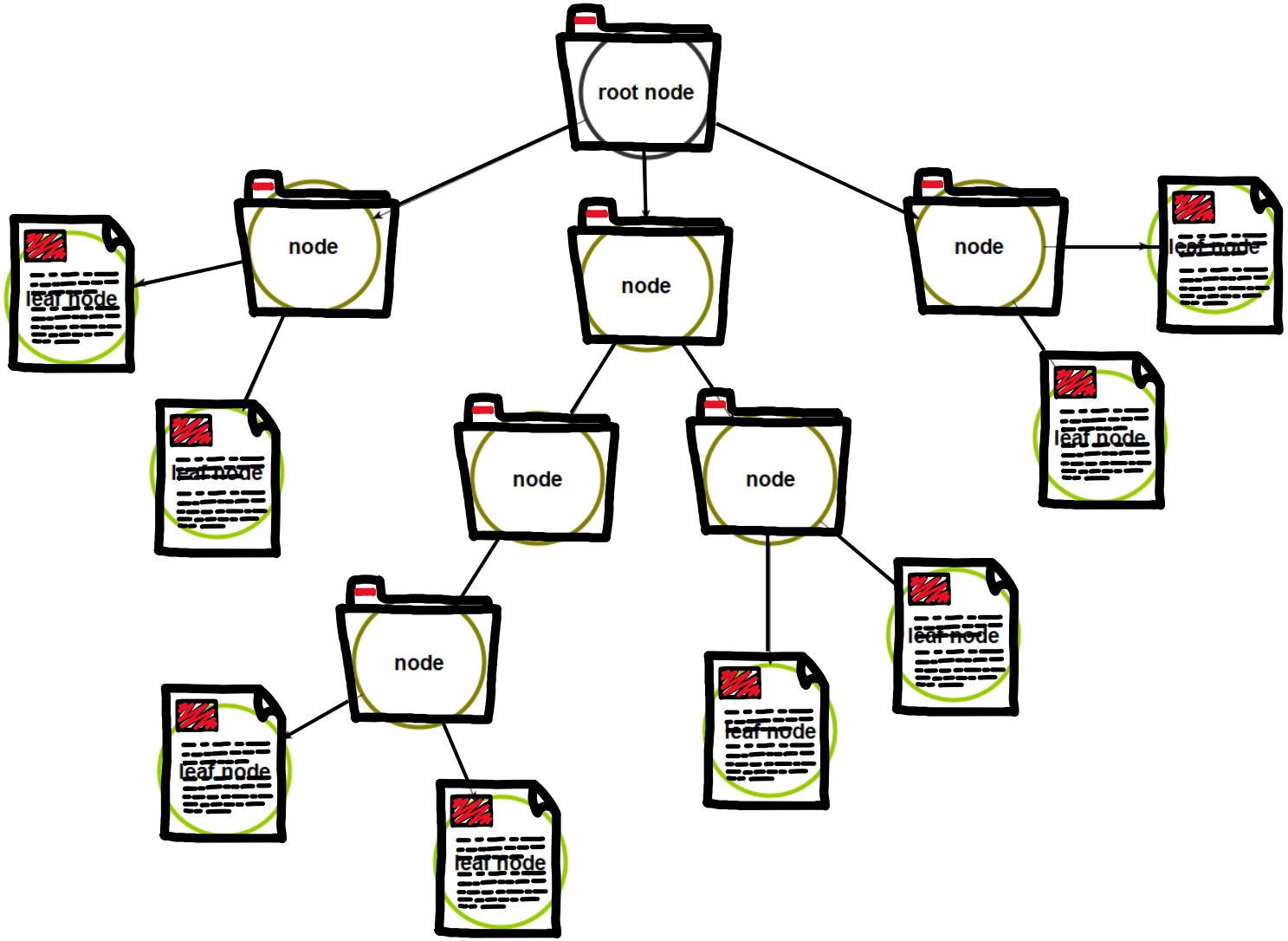
Tree


Parent
node
node
node
node
leaf node
leaf node
leaf node
leaf node
Decendants

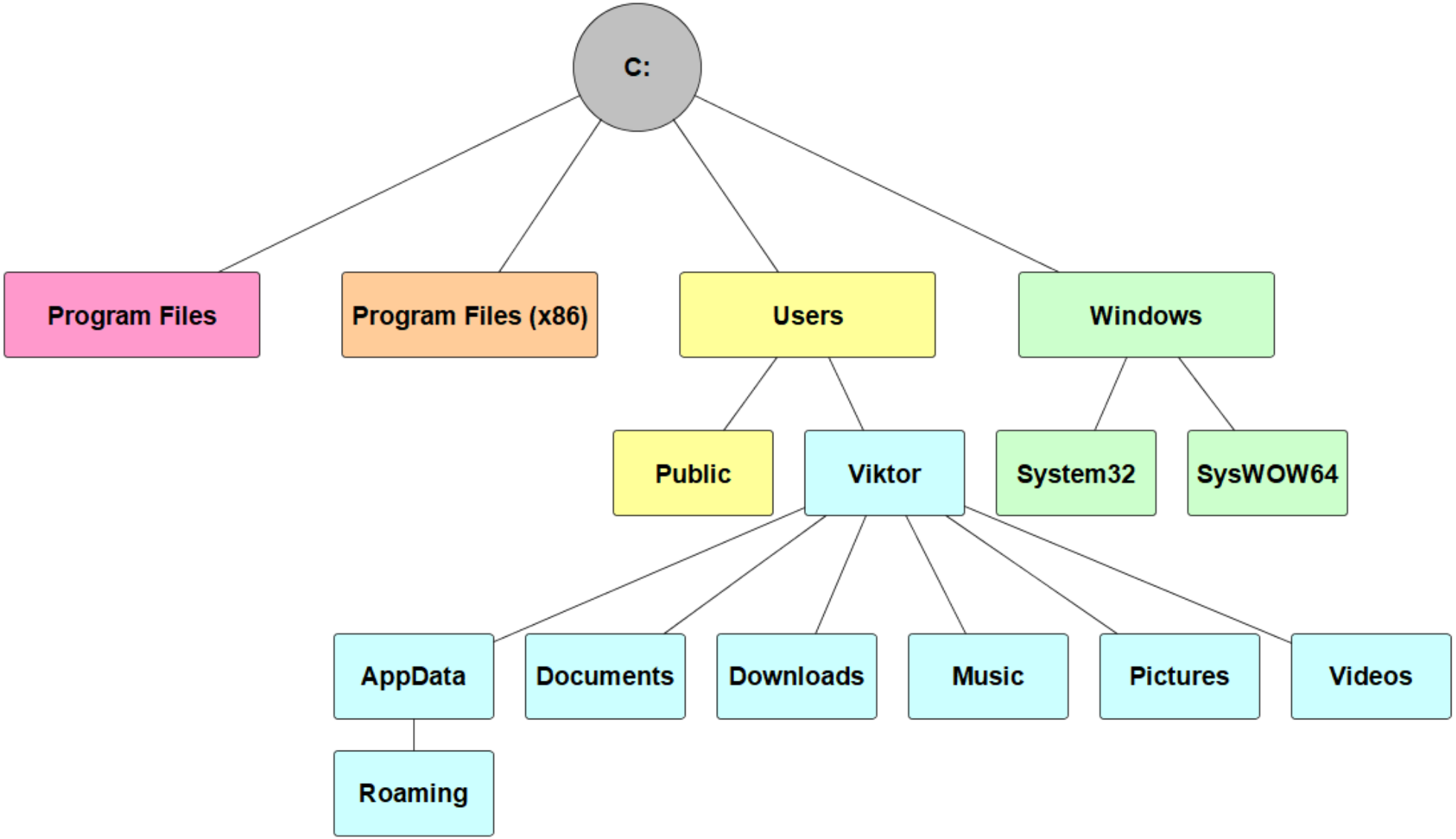
# 4. Filstruktur

# Filhierarki

Filsystem är en hirerkisk strukturer av *filer* och *kataloger*



# Windows filstruktur



# Windows filstruktur

| C: | File system root (boot volume) |
|---|---|
| **+- Program Files** | 64-bit programs |
| **+- Program Files (x86)** | 32-bit programs |
| **+- Users** | Users home folders |
| **\|  +- Public** | Files shared between users |
| **\|  +- Viktor** | A user's home folder |
| **\|  +- AppData** | User specific app settings |
| **+- Windows** | Windows & library files |
| **+- System32** | 64-bit DLL-libraries |
| **+- SysWOW64** | 32-bit DLL-libraries |

# **Windows filsökvägar**

Windows File Paths

1. **Absolut-** eller **fullständig filsökväg**
   utgår ifrån roten av filsystemet

2. **Relativfilsökväg**
   utgår ifrån en katalog du befinner dig i

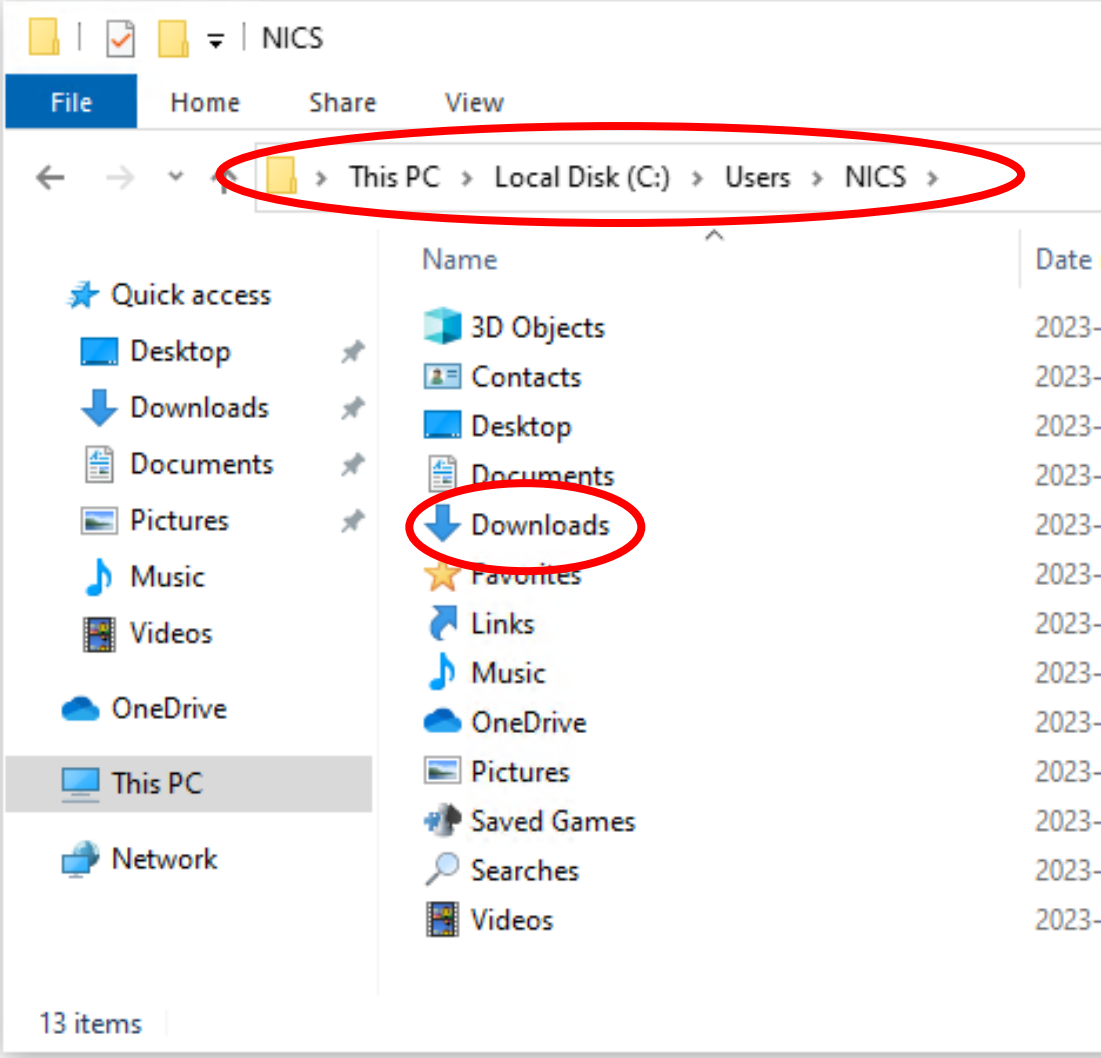
# Windows filsökvägar

## Windows File Paths

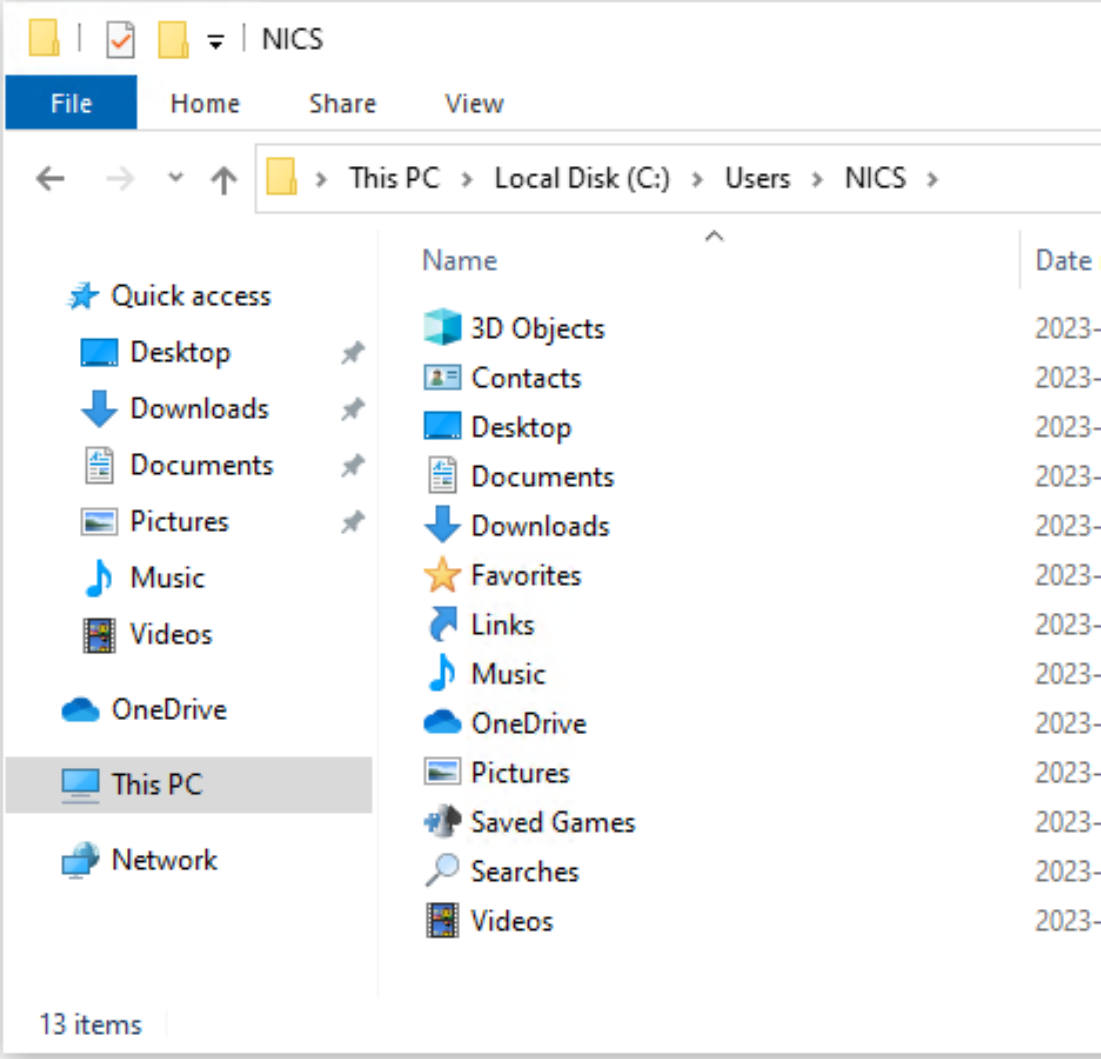
Ex.

C:\Users\NICS\Downloads

Downloads



# Navigering med GUI



# 5. Navigering med CLI

# Navigering med CLI

Kan inte gissa sig fram på samma sätt som med ett GUI

Använder kommandon

| | |
|---|---|
| **dir** | Lists contents of directories |
| **cd** | Change current directory |

# directory

Notera prompten

```
C:\Users\NICS> dir

 Directory of C:\Users\NICS

2023-08-24  10:14    <DIR>          .
2023-08-24  10:14    <DIR>          ..
2023-08-30  16:00    <DIR>          Documents
2023-08-29  08:17    <DIR>          Downloads
2023-08-24  10:14    <DIR>          Music

C:\Users\NICS>
```

# directory

```
C:\Users\NICS> dir Downloads

 Directory of C:\Users\NICS\Downloads

2023-08-30  08:17    <DIR>          .
2023-08-30  08:17    <DIR>          ..
2023-08-29  16:04           398 720 Firefox Installer.exe
2023-08-30  08:17    <DIR>          ProcessExplorer


C:\Users\NICS>
```

# change directory

## Absolutsökväg

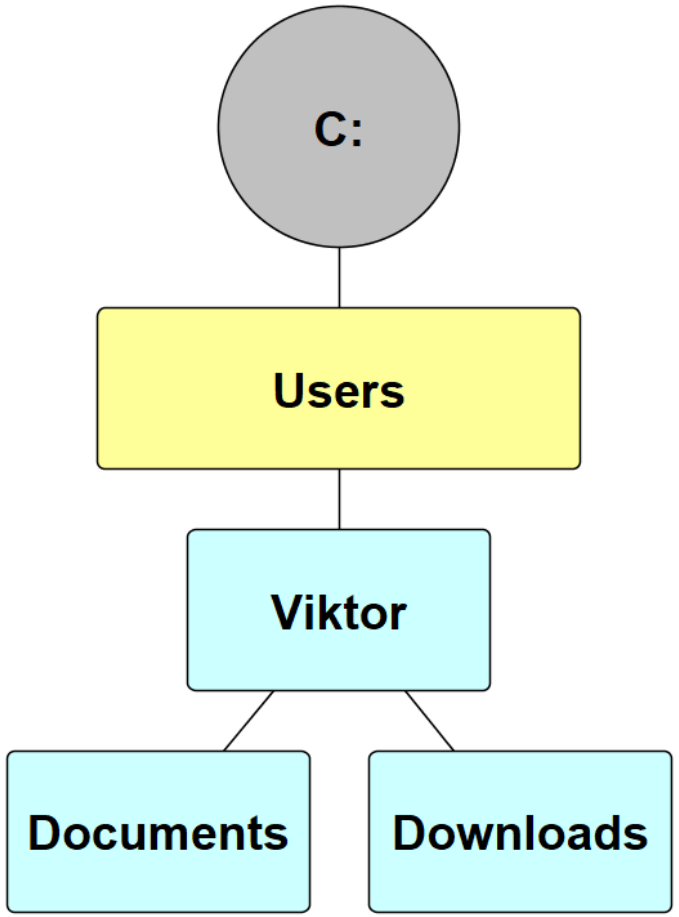
```
C:\Users\NICS> cd C:\Users\NICS\Downloads

C:\Users\NICS\Downloads>
```

## Relativsökväg

```
C:\Users\NICS> cd Downloads

C:\Users\NICS\Downloads>
```



# Hur går man tillbaka i ett CLI?

## Om jag befinner mig:

```
C:\Users\NICS\Downloads>
```

## Absolutsökväg:

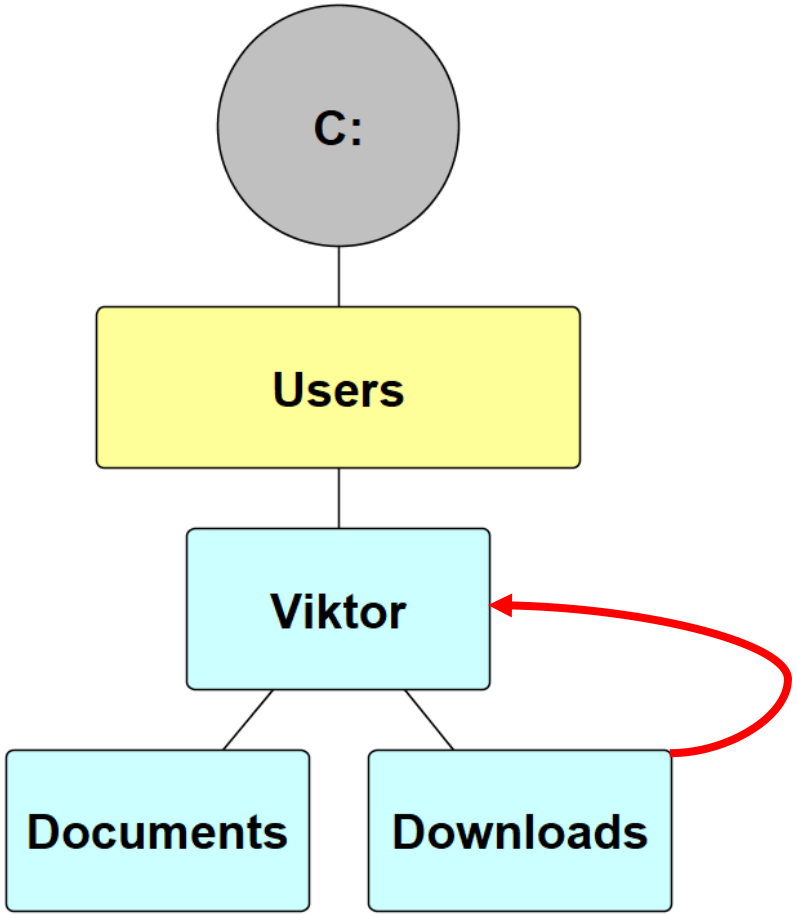
```
C:\Users\NICS\Downloads>cd C:\Users\NICS

C:\Users\NICS>
```

## Relativsökväg:

```
C:\Users\NICS\Downloads> cd ..

C:\Users\NICS>
```



# Specialtecken

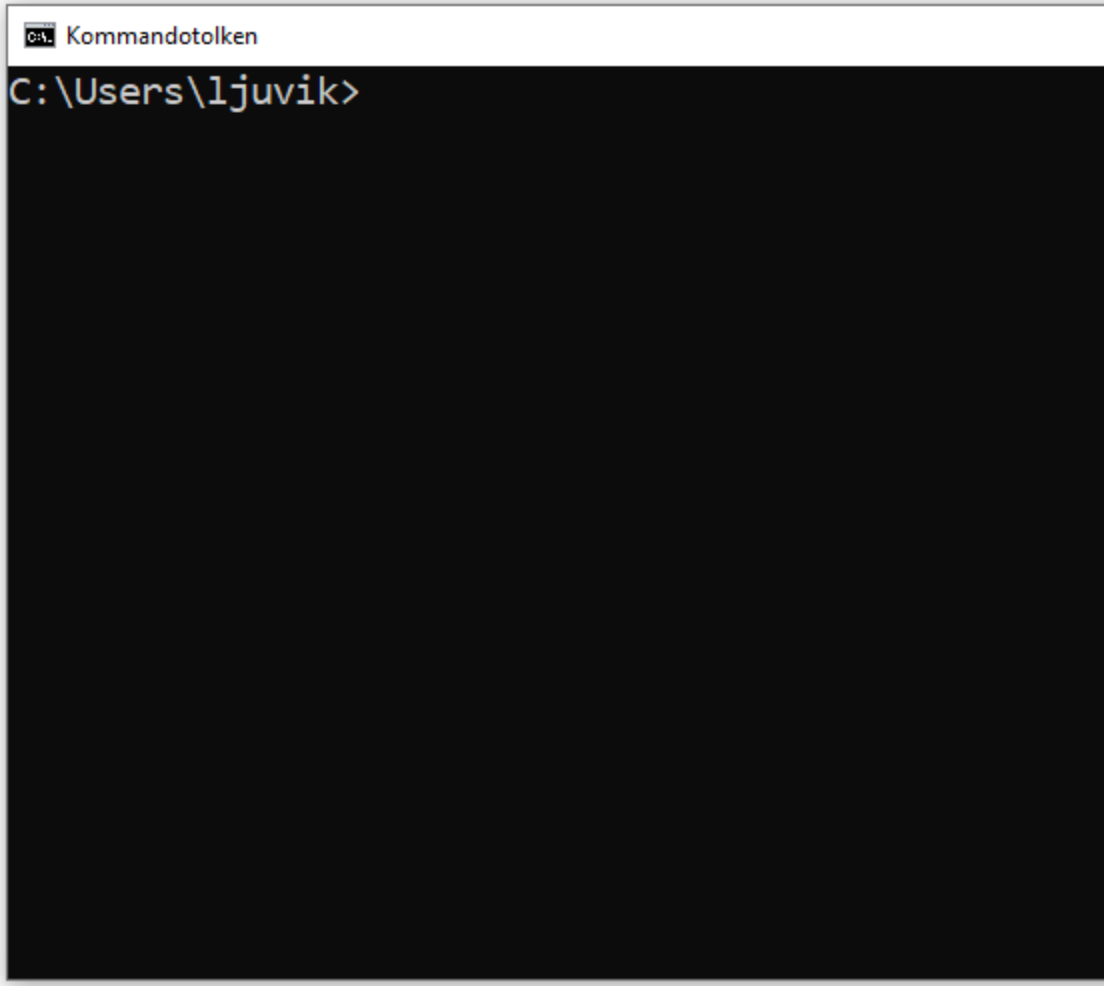
| | |
|---|---|
| . | Current directory |
| .. | Parent directory |

```
C:\Users\NICS> dir

 Directory of C:\Users\NICS

2023-08-24  10:14    <DIR>          .
2023-08-24  10:14    <DIR>          ..
2023-08-30  16:00    <DIR>          Documents
```

# Navigering med CLI



# 6. Filsystem

# Allocation Unit

Den enhet lagringsyta som filsystemet kan allokera

’Det minsta filsystemet kan hantera’

Kallas oftast:

- **Block size**
- **Cluster size**

# Allocation Unit

Ex.

Block size: 4 KiB (4096 Bytes)

Filstorlek: 3 kB  (3000 Bytes)

Hur mycket plats kommer filen att använda?

➔ Ett block, alltså 4 KiB

# Allocation Unit

Ex.

Block size: 4 KiB (4096 Bytes)

Filstorlek: 6 kB  (6000 Bytes)

Hur mycket plats kommer filen att använda?

➔ två block, alltså 8 KiB

# Filsystem

| | |
|---|---|
| **APFS** | macOS, iOS, iPadOS, watchOS |
| **FAT** | ’portable lagring’ |
| **EXT4** | Linux |
| **NTFS** | Windows |
| **ReFS** | Windows |
| **UFS** | UNIX & BSD |

# **F**ile **A**llocation **T**able

Alla vanliga operativsystem kan använda **FAT**

Används därför ofta för **USB flash drives**

*A SanDisk-brand USB thumb drive, SanDisk Cruzer Micro, 4GB*, by Evan-Amos, 2019, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=11294374). Public domain.

# FAT32

Introducerades med *Windows 95* och är idag den vanligaste varianten av **FAT**

- max 2 TiB volume
- max 4 GiB file size
- no ’advanced features’
  - file permissions, encryption, compression…
- no journaling

# exFAT

Utökar **FAT32** för att stödja mer lagring och större filer

- max 128 PiB volume and file size

Används primärt för **SD-kort** av SDXC-format

# **NT F**ile **S**ystem

Med *Windows NT* kom filsystemet **NTFS**

Standard filsystemet för moderna Windows

- max 16 TiB volume and file size
- journaling
- ’advanced features’
  - file permissions, compression, resizing, quotas, volume shadow copy…

# **Re**silient **F**ile **S**ystem

Hittills inte särskilt populärt men kommer antagligen ersätta **NTFS** på sikt

Fokus på tillförlitlighet

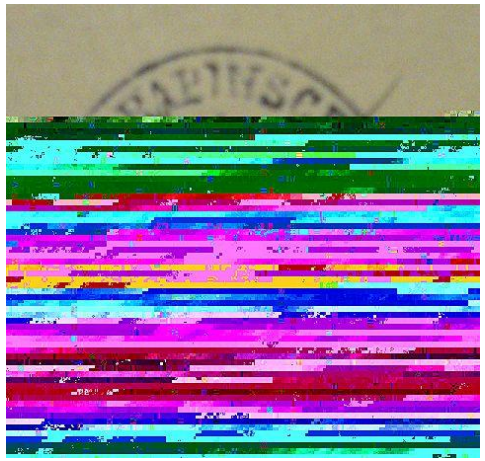
- storage spaces
- checksum based validation
- read/write data validation
- scheduled corruption scrubbing

# Journaling Filesystems

Använder en så kallad **journal**

Där filsystemsändringar sparas i sekundärminnet innan de genomförs

Vilket tillåter filsystemet att korrigera vanliga fel som annars hade lett till datakorruption



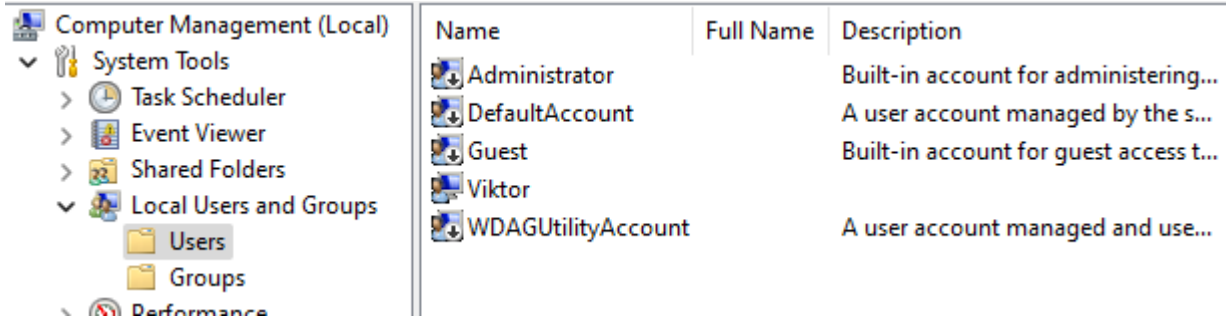
*Result of an almost completely failed data restoration of a JPEG photograph on a hard drive. The file was overwritten after about a year with other data and is visible only to a small extent in the original,* by Mattes, 2008, Wikimedia (https://commons.wikimedia.org/w/index.php?curid=5217861). Public domain.

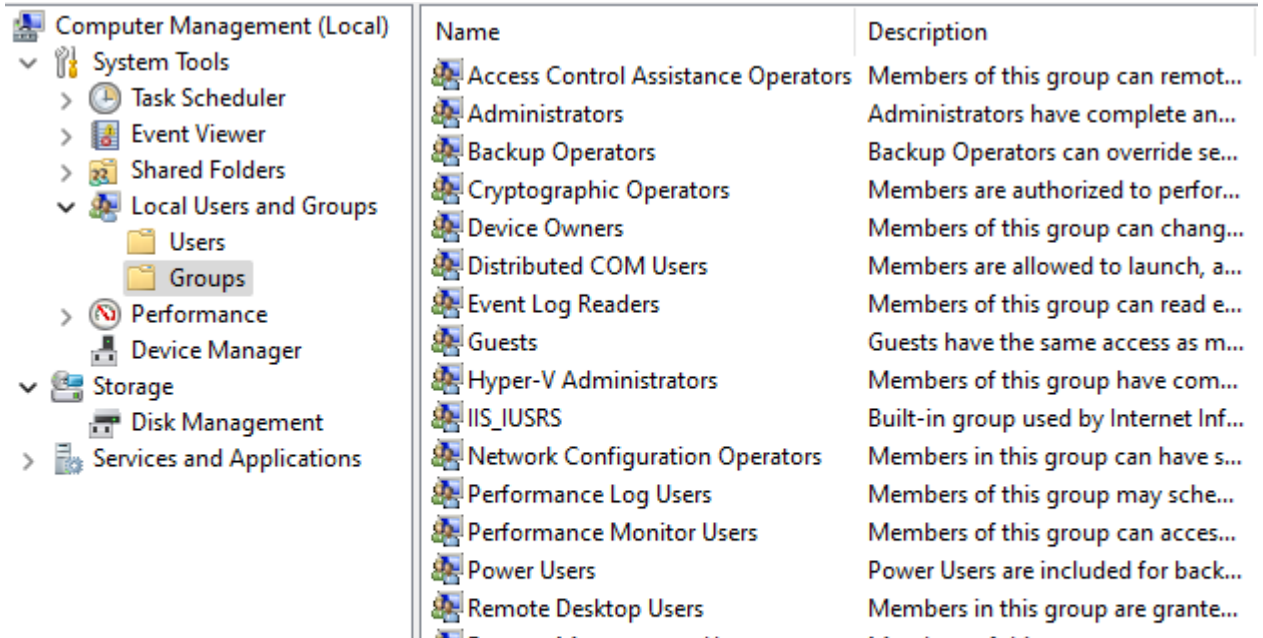
# 7. NTFS File Permisisons

# Users and Groups

## Användare och grupper

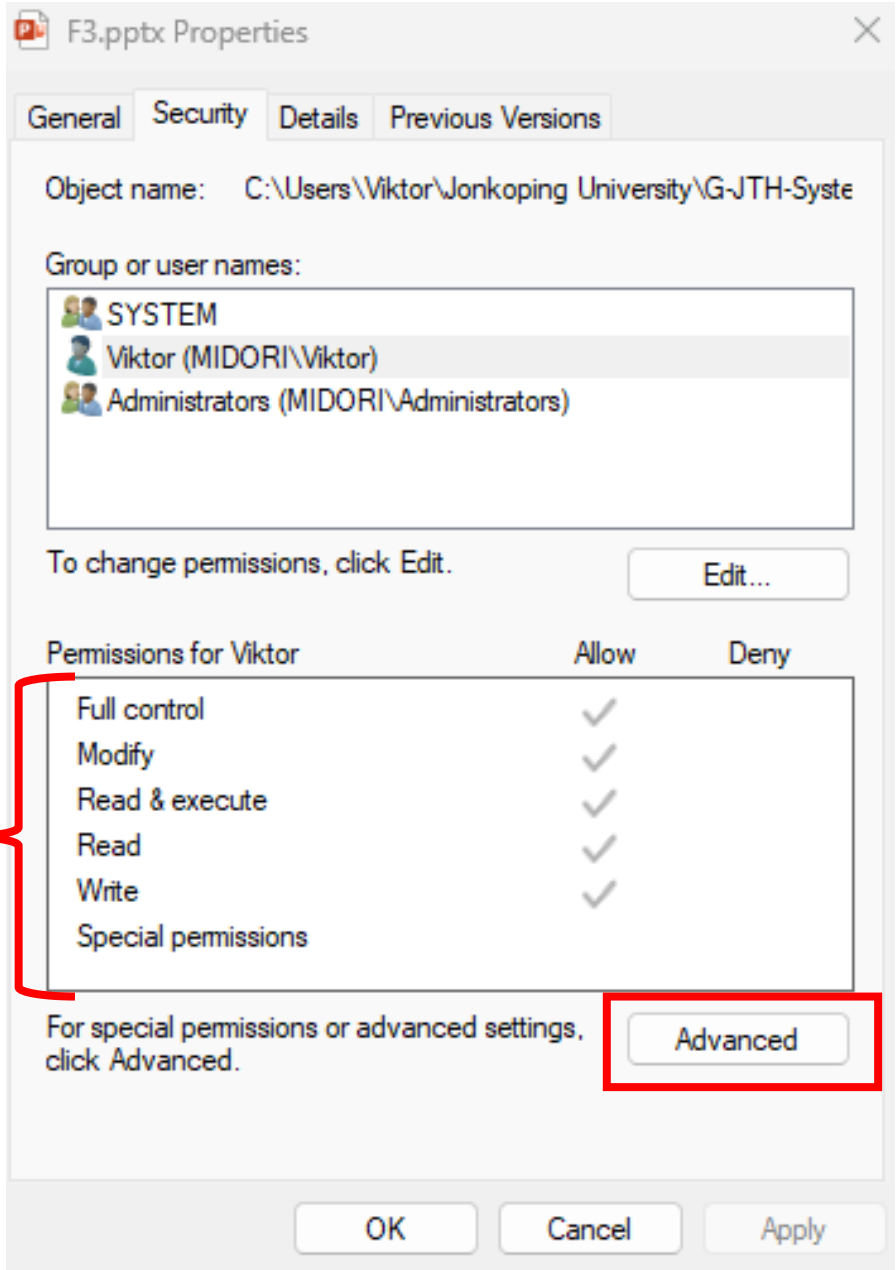
Användare och/eller grupper kan tilldelas rättigheter till filer eller kataloger



# NTFS File Permissions

## NTFS filrättigheter



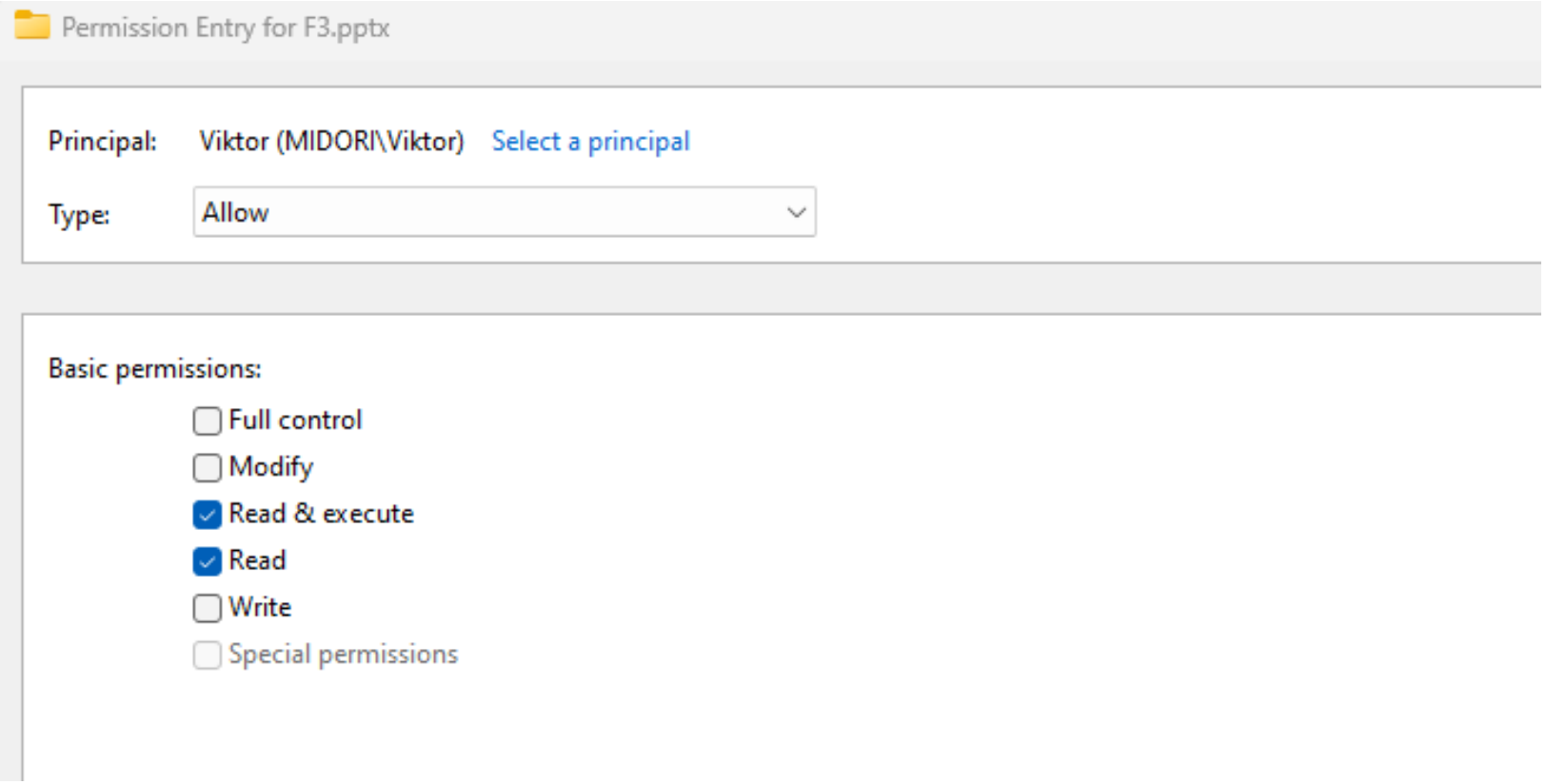

Rättigheter

Använd *Advanced Security Settings*!

# NTFS File Permissions

NTFS filrättigheter

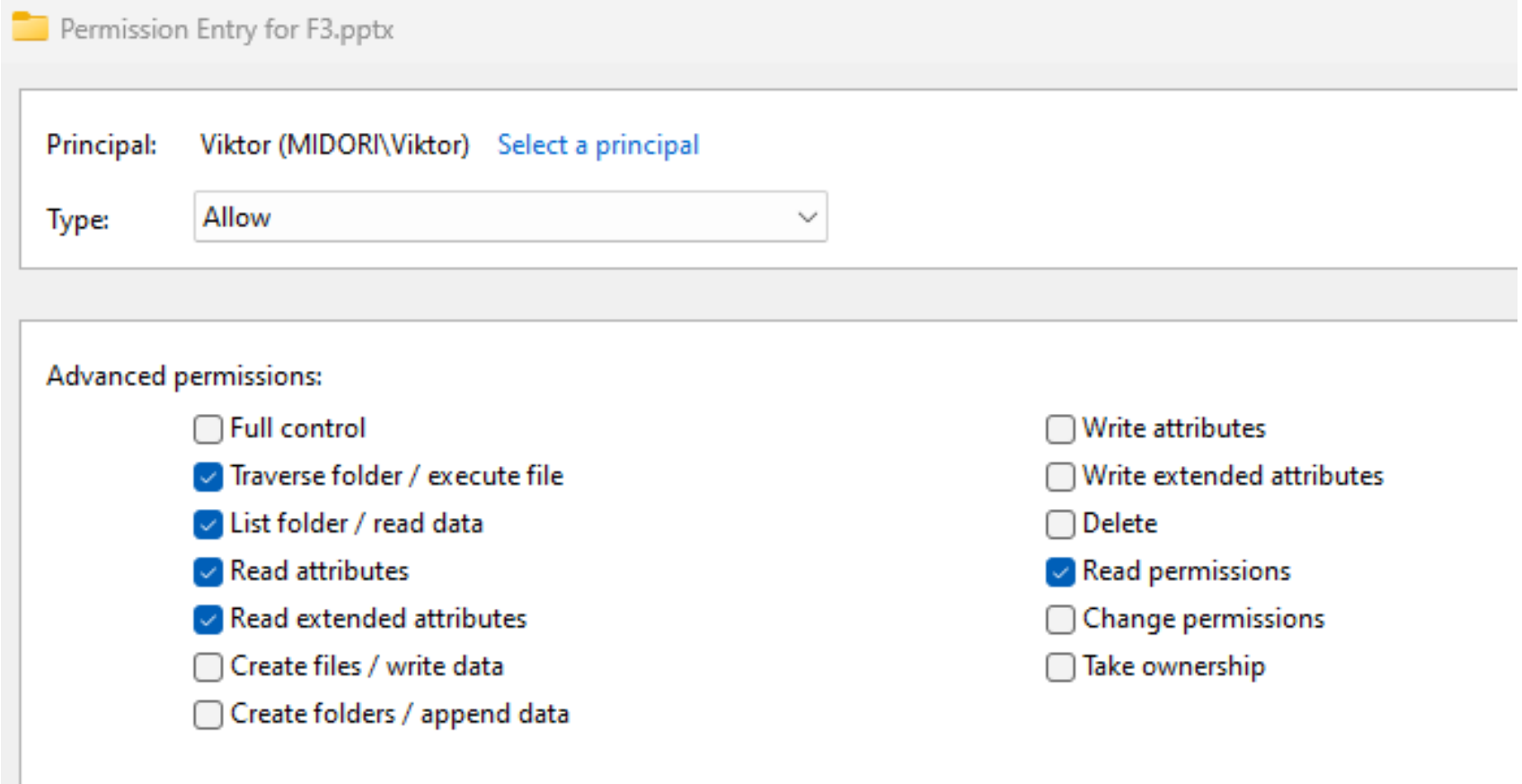
## Basic permissions (molecular)



# NTFS File Permissions

NTFS filrättigheter

## Advanced permissions (atomic)



# NTFS Ownership

NTFS ägandeskap

- Filer och kataloger har en ägare, antingen en användare eller grupp
- Ägaren kan tilldela sig själv och andra rättigheter

# NTFS Take Ownership

Rättigheten **take ownership** (atomic) ger andra förmågan att ta över ägandeskapet

**Administrator** kan alltid ta över ägandeskapet, oavsett tilldelade rättigheter

# NTFS File Permissions

## NTFS filrättigheter

| Permissions name | Ability granted |
|---|---|
| **Read** | View contents, permissions and attributes for a file or folder |
| **Write** | Add to file contents<br>Create files/folders in folder |
| **Read & Execute** | **Read** + run executable files |
| **Modify** | **Read & Execute** + **Write** and can delete file/folder |

# NTFS File Permissions

## NTFS filrättigheter

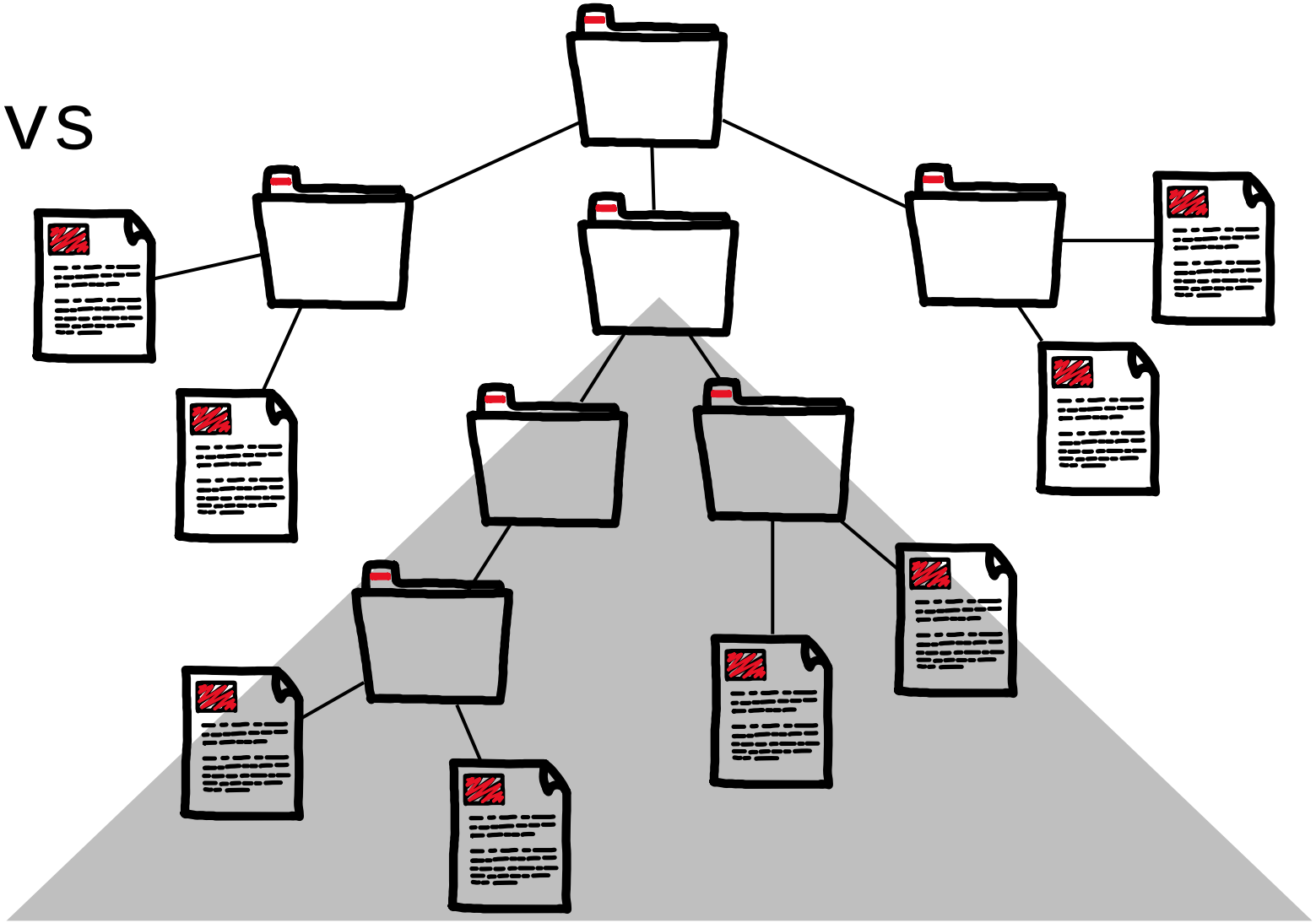
| Permissions name | Ability granted |
|---|---|
| **Full Control** | All permissions and the ability to take ownership |
| **List Folder Contents** | Only applies to folders; ability to view contents of folder |
| **Special Permissions** | Any customized grouping of advanced permissions (atomic) |

# NTFS Implicit Deny

Finns det inga tilldelade rättigheter, har man inga!

# NTFS Inheritance

Rättigheter ärvs
från *parent*



# NTFS Disable Inheritance?

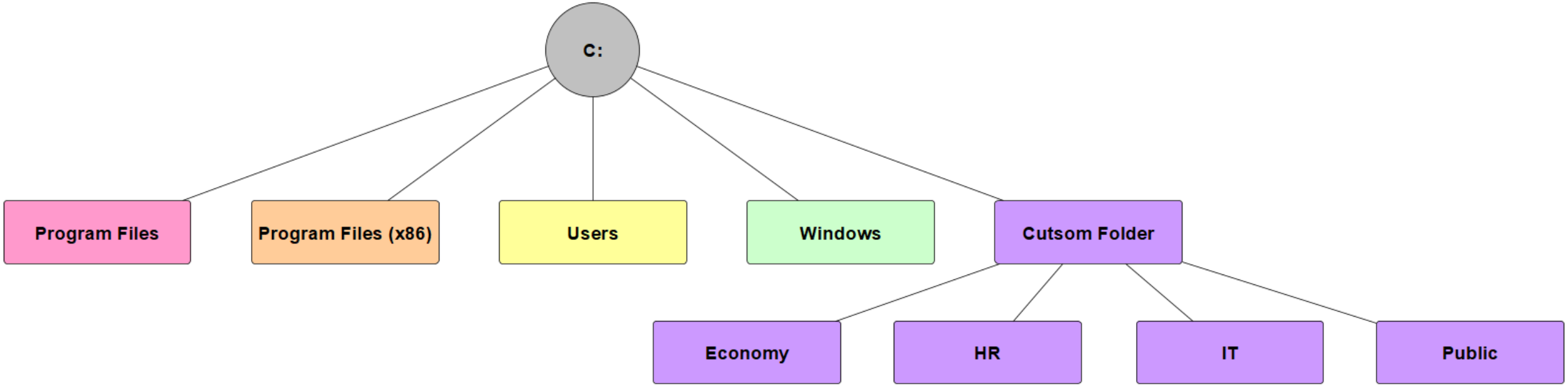
Du kan bryta rättighets-arvet…

Använd dig utav arv!

- Låt standard filstrukturen vara
- För ’egen definierad’ filstruktur bör arvet brytas, men låt *child nodes* ärva rättigheter

# NTFS Disable Inheritance?

Ex.



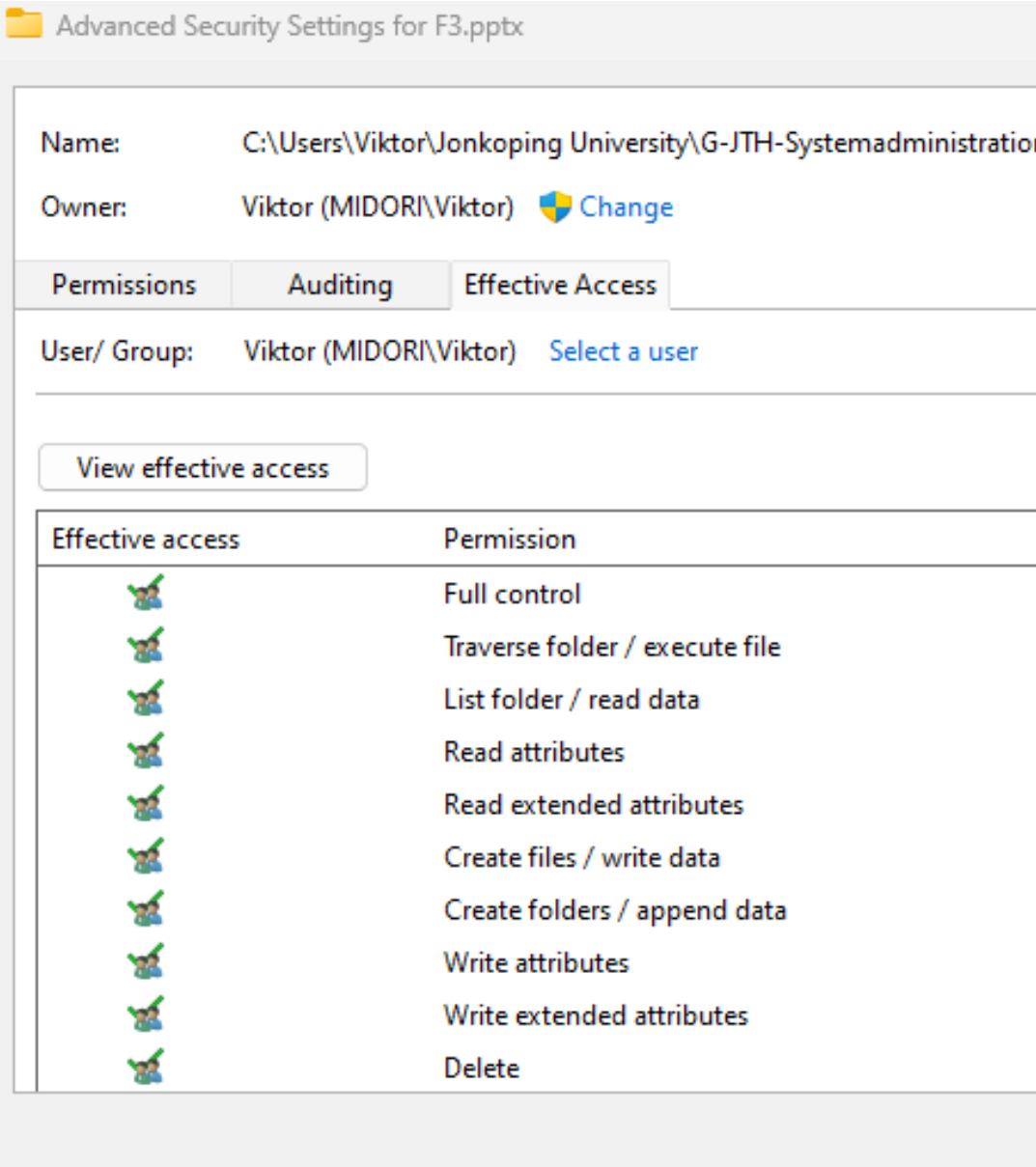
# NTFS Effective Permissions

Det kan bli rörigt med flera arv & manuellt satta rättigheter…

- **deny** trumfar **allow**
- **explicit permissions** trumfar **inherited permissions**

# NTFS Effective Permissions

1. Explicit Deny
2. Explicit Allow
3. Inherited Deny
4. Inherited Allow

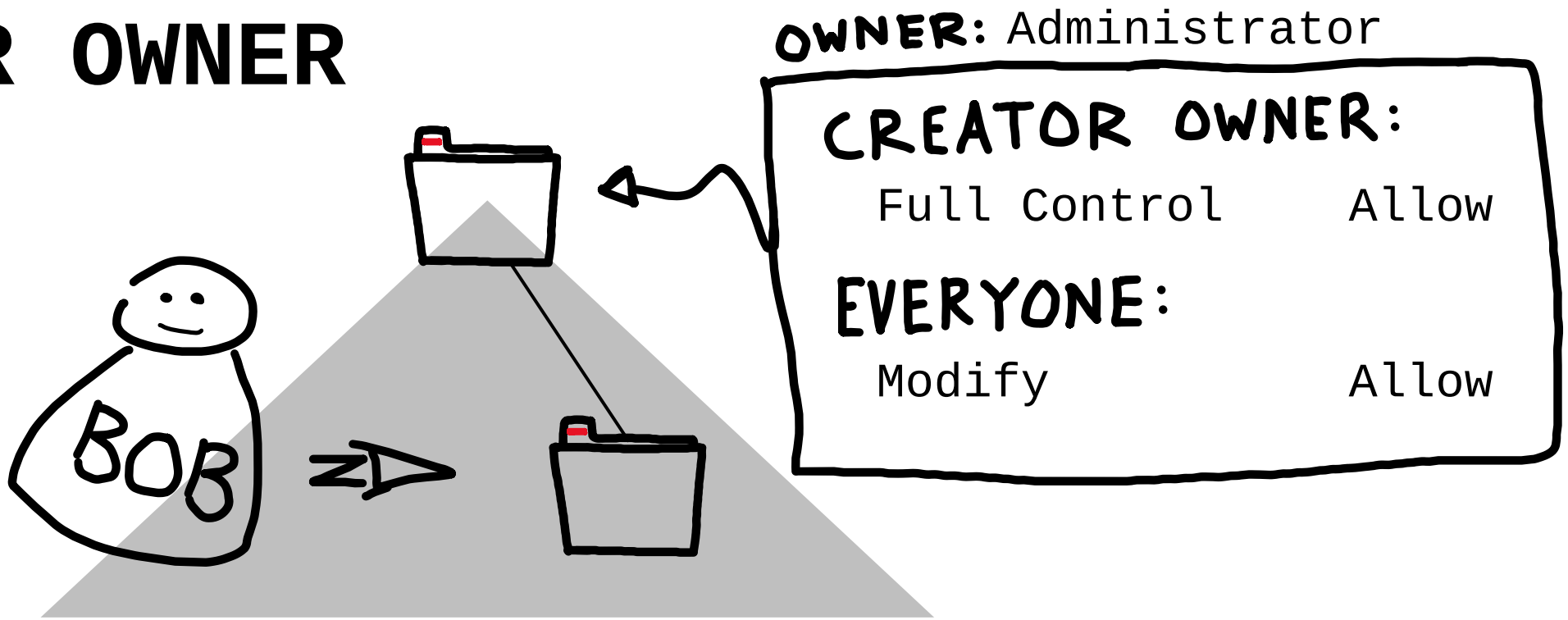


# Special Identity Groups

| Group name | Members |
|---|---|
| **Authenticated Users** | All interactive user accounts |
| **Everyone** | All interactive users<br>+ guests (default disabled)<br>+ service accounts (background processes) |
| **CREATOR OWNER** | Functions like a template |

# CREATOR OWNER

Ex.

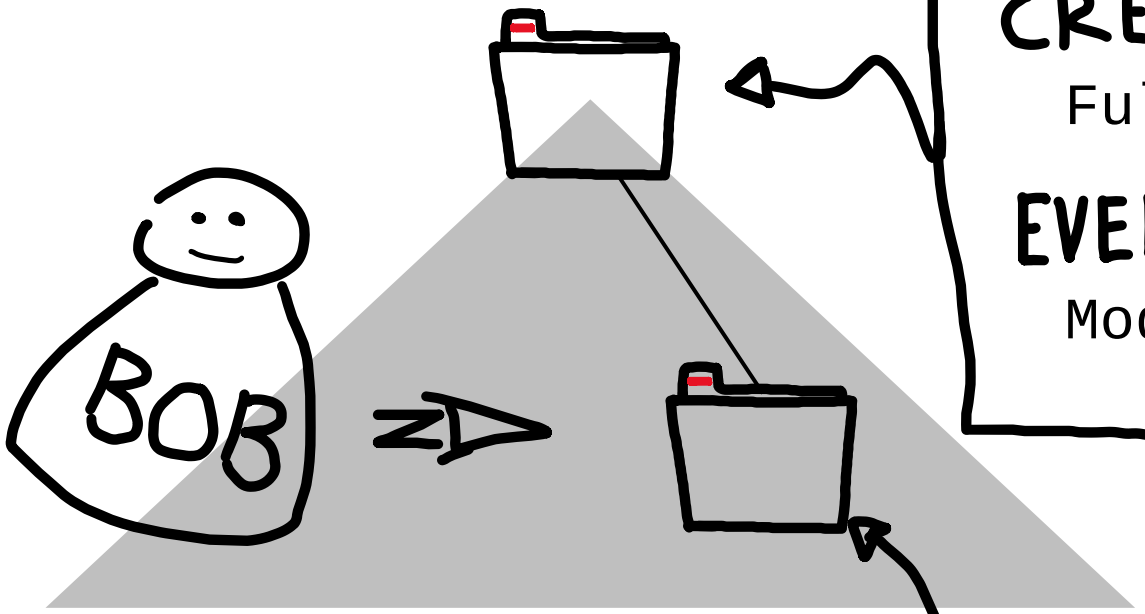



Inheritance

| | |
|---|---|
| **Modify** | **Read & Execute** + **Write** and can delete file/folder |
| **Write** | Add to file contents<br>Create files/folders in folder |

# CREATOR OWNER

Ex.




Inheritance

OWNER: Administrator

CREATOR OWNER:

| | |
|---|---|
| Full Control | Allow |

EVERYONE:

| | |
|---|---|
| Modify | Allow |

OWNER: Bob

BOB:

| | |
|---|---|
| Full Control | Allow |

# Till nästa gång

- Läs kurslitteraturen
- Onsdag: föreläsning

Tack för idag